

Day 3: Configuring Basic Network Hardware

Theory: IP Addressing Basics and Network Configuration

Practical: Set up a router with default settings

Status: Completed / Ready for Submission

Objective

To understand basic IP addressing, identify the network details assigned by DHCP, configure a static IP address for testing, and understand the router's DHCP range.

Task 1: Router Admin Panel Login

Answer: The router admin panel can usually be opened by entering the router's Default Gateway address in a web browser. Common addresses are **192.168.0.1** or **192.168.1.1**.

Example: Open a browser → enter **192.168.1.1** → router login page appears → enter the router's admin username/password.

Screenshot: Insert a screenshot of your router login page here.

Task 2: IP Address, Subnet Mask and Default Gateway

Network Detail	Example Value
IP Address	192.168.1.50
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.1 (example)

Note: These are sample values. For final submission, replace them with the actual values shown on the connected laptop/mobile device.

Task 3: Static IP Configuration and Testing

Answer: The device can be changed from DHCP (Obtain IP automatically) to a static IP within the same network range. For example, if the router is 192.168.1.1, a suitable test address can be 192.168.1.50, with subnet mask 255.255.255.0 and gateway 192.168.1.1.

Setting	Example
IP Address	192.168.1.50
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
Preferred DNS	8.8.8.8

Test Result (example): Internet access works when the static IP is correctly configured and the address does not conflict with another device.

Important: After testing, change the setting back to **Obtain IP automatically (DHCP)** to avoid network connectivity problems.

Task 4: DHCP Range and Duplicate IP Address

Example DHCP Range: 192.168.1.100 – 192.168.1.200

Explanation: DHCP automatically assigns IP addresses to devices connected to the network. If two devices use the same IP address, an IP conflict can occur. This may cause one or both devices to lose network/internet connectivity or communicate incorrectly until the duplicate address is fixed.

Conclusion

In this practical, I learned how a router's Default Gateway is used to access its admin panel, how to identify IP address, subnet mask and gateway, how DHCP assigns IP addresses, and how a static IP can be configured temporarily for testing. I also learned why duplicate IP addresses can create network conflicts.

GitHub Submission Notes

- Add the router login-page screenshot under Task 1.
- Replace the sample network values in Task 2 with actual values from your device.
- Add a screenshot of the static-IP settings/test if required by the instructor.
- Add a screenshot of the router's DHCP range under Task 4.
- Save the final PDF and upload it to the GitHub repository.